

Reason with number properties

Use the scaffold, show each step and ask for an adult check when marked.

Name: _____ Date: _____

1. Read the model: Find the smallest number divisible by 6, 8 and 9. Copy or complete the first step.

2. A rectangle has area 96 square units and whole-number sides. Give every possible side pair.

3. Which prime factors make 84?

4. Miro says: Miro assumes every odd number is prime. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: Find the smallest number divisible by 6, 8 and 9. Copy or complete the first step.
Answer 72.
2. A rectangle has area 96 square units and whole-number sides. Give every possible side pair.
Answer 1 x 96, 2 x 48, 3 x 32, 4 x 24, 6 x 16 and 8 x 12.
3. Which prime factors make 84?
Answer 2 x 2 x 3 x 7.
4. Miro says: Miro assumes every odd number is prime. Is this correct? Explain.
Answer Odd composite numbers have factors other than 1 and themselves, for example 9 and 15.

Reason with number properties

Work independently. Show a complete method and check at least one answer.

Name: _____ Date: _____

1. Find the smallest number divisible by 6, 8 and 9.

6. Check this result using an inverse, estimate or known fact: Which prime factors make 84?

2. A rectangle has area 96 square units and whole-number sides. Give every possible side pair.

7. Find two prime numbers with a sum of 60 and a difference of 14.

3. Which prime factors make 84?

8. Identify and correct this misconception: Miro assumes every odd number is prime.

4. Solve this independently and show every stage: Find the smallest number divisible by 6, 8 and 9.

9. Write a complete sentence explaining how you checked one answer.

5. Use a second method or representation for: A rectangle has area 96 square units and whole-number sides. Give every possible side pair.

10. Write and solve a similar question to: Which prime factors make 84?

Teacher answers and checking prompts

1. Find the smallest number divisible by 6, 8 and 9.
Answer 72.
2. A rectangle has area 96 square units and whole-number sides. Give every possible side pair.
Answer 1 x 96, 2 x 48, 3 x 32, 4 x 24, 6 x 16 and 8 x 12.
3. Which prime factors make 84?
Answer $2 \times 2 \times 3 \times 7$.
4. Solve this independently and show every stage: Find the smallest number divisible by 6, 8 and 9.
Answer 72.
5. Use a second method or representation for: A rectangle has area 96 square units and whole-number sides. Give every possible side pair.
Answer Expected result: 1 x 96, 2 x 48, 3 x 32, 4 x 24, 6 x 16 and 8 x 12. Method or representation may vary.
6. Check this result using an inverse, estimate or known fact: Which prime factors make 84?
Answer Expected result: $2 \times 2 \times 3 \times 7$. A valid check must be shown.
7. Find two prime numbers with a sum of 60 and a difference of 14.
Answer 23 and 37.
8. Identify and correct this misconception: Miro assumes every odd number is prime.
Answer Odd composite numbers have factors other than 1 and themselves, for example 9 and 15.
9. Write a complete sentence explaining how you checked one answer.
Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.
10. Write and solve a similar question to: Which prime factors make 84?
Answer Answers vary. The new question must use the same mathematical structure and include a correct solution.

Reason with number properties

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: _____ Date: _____

1. Find two prime numbers with a sum of 60 and a difference of 14.

6. Create a new example that tests the same idea as: Which prime factors make 84?

2. Prove the answer to this question, rather than only calculating it: Find the smallest number divisible by 6, 8 and 9.

7. Find a second method or representation. Compare its efficiency with your first method.

3. Solve this in two different ways and compare them: A rectangle has area 96 square units and whole-number sides. Give every possible side pair.

8. Write one always, sometimes or never statement about today's learning and justify it.

4. Work backwards from the answer to reconstruct the question: $2 \times 2 \times 3 \times 7$.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.

5. Explain why this reasoning fails: Miro assumes every odd number is prime.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.

Teacher answers and checking prompts

1. Find two prime numbers with a sum of 60 and a difference of 14.
Answer 23 and 37.
2. Prove the answer to this question, rather than only calculating it: Find the smallest number divisible by 6, 8 and 9.
Answer Expected result: 72. Proof or justification must match the lesson structure.
3. Solve this in two different ways and compare them: A rectangle has area 96 square units and whole-number sides. Give every possible side pair.
Answer Expected result: 1 x 96, 2 x 48, 3 x 32, 4 x 24, 6 x 16 and 8 x 12. Two valid methods and a comparison are required.
4. Work backwards from the answer to reconstruct the question: $2 \times 2 \times 3 \times 7$.
Answer One valid reconstruction is: Which prime factors make 84? Other equivalent questions are acceptable.
5. Explain why this reasoning fails: Miro assumes every odd number is prime.
Answer Odd composite numbers have factors other than 1 and themselves, for example 9 and 15.
6. Create a new example that tests the same idea as: Which prime factors make 84?
Answer Answers vary. The example and solution must preserve the mathematical structure.
7. Find a second method or representation. Compare its efficiency with your first method.
Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.
8. Write one always, sometimes or never statement about today's learning and justify it.
Answer Answers vary. A valid example and counterexample should support the classification.
9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.
Answer Answers vary. The error must be mathematically relevant and the correction must be precise.
10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.
Answer Answers vary. The explanation must distinguish the mathematical structure from the changed value.